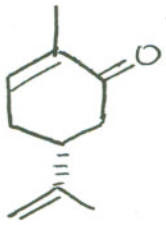
ORGANIC CUME
NOVEMBER 16, 2001

The following problems are from Srikrishna, et al. *Tetrahedron Lett.* **2000**, *41*, 6643-7.

(1) Draw the structures of intermediates A-N. Be sure that stereochemistry is unambiguously indicated where appropriate. Also be sure that your structures have reasonable bond angles.

(2) Draw the mechanism of ***one*** of the following steps: 4, 6, 8, 18.



1. (i) LDA, THF
   (ii) MeI, 0°C → rt
2. repeat

*3. MeMgBr, Et_2O, 0°C → rt

A

4. PCC, silica gel, CH_2Cl_2, rt

B

5. NBS, CH_2Cl_2, MeOH, 0°C → rt

C

6. tBuOK, tBuOH, THF, −5°C → rt

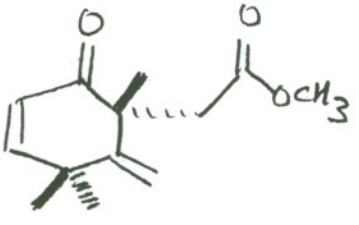
D

7. O_3, O_2, MeOH, CH_2Cl_2, $NaHCO_3$, −70°C

E

8. Ac_2O, NEt_3, DMAP, C_6H_6, reflux

F



G

G

$\xrightarrow[\text{EtOAc}]{\text{9. 5\% Pd/C, } H_2}$

$\xrightarrow[\text{MeOH, } H_2O \text{ 1:1, reflux}]{\text{10. 5\% NaOH}}$

H

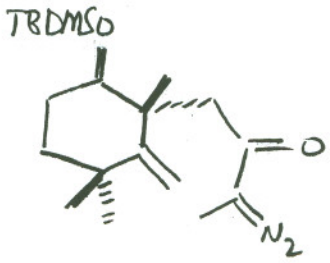
$\xrightarrow[\text{0°C}]{\text{11. } NaBH_4 \text{ THF}}$

$\xrightarrow[\text{Et}_2\text{O 0°C}]{\text{12. } CH_2N_2}$

I

J

$\xrightarrow{\text{13. – 16.}}$



$\xrightarrow[\text{hν}]{\text{17. Cu, CuSO}_4 \text{ cyclohexane, reflux}}$

K

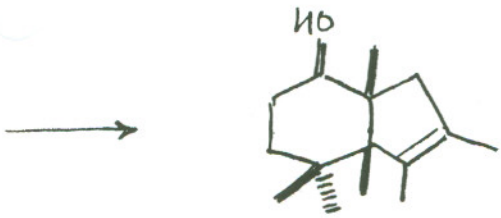
L

$\xrightarrow[\text{tBuOH -33°C}]{\text{18. Li}^0 \text{ NH}_{3(l)}}$

$\xrightarrow{\text{19. – 21.}}$

M

N



(−)-thaps-8-en-5-ol

Draw the mechanism of <u>one</u> of the following steps: 4, 6, 8, 18.